
A Comparative Study of Estimation Algorithms for the Latent Block Model

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Abstract

Model-based clustering methods are effective in low-dimensional settings, but they often fail in high-dimensional contexts due to both statistical and computational limitations. To address these issues, we consider the Latent Block Model (LBM), a probabilistic co-clustering framework that can handle high-dimensional data by clustering individuals and features. In this project, we aim to study and compare different algorithms used to estimate the parameters of the LBM, with a focus on their theoretical assumptions, statistical properties, and empirical performance.

1 Introduction

Model-based clustering methods such as Gaussian mixture models or stochastic block models have been used to uncover latent structures in data. However, these approaches face important limitations when dealing with high-dimensional data, where the number of variables can exceed the number of observations. In such cases, parameter estimation becomes statistically unstable and computationally challenging.

The Latent Block Model (LBM) offers a natural solution to this problem by performing co-clustering, i.e. simultaneously clustering individuals and variables (or individuals). This dual structure reduces the effective dimensionality and leads to more interpretable results. Despite the popularity of the LBM, the choice of estimation algorithm has a strong impact on both the accuracy and the feasibility of the inference. Different approaches, such as Expectation-Maximization (EM), Variational EM (VEM), and Markov Chain Monte Carlo (MCMC), rely on distinct statistical principles and come with different trade-offs in terms of bias, variance, convergence properties, and computational cost. So we seek to compare these algorithms to determine why we might prefer one over another.

Thus, the main problematic of this project is :

How do different estimation algorithms for the Latent Block Model compare in terms of their theoretical assumptions, statistical properties, and empirical performance ?

2 Background & Related Grounds

2.1 Mixture Models

The general model-based approach assumes that for each individual i , there is an unknown (or latent) label Z_i that can take a finite number of values from $\{1, \dots, K\}$. The distribution of the observed variables Y_i depends on the value of this latent variable Z_i .

The simple mixture model assumes that the observations are independent and follow a parametric distribution :

- The latent variables (Z_i) are i.i.d. with $P(Z_i = k) = \pi_k$;

- The observed variables (Y_i) are independent conditional on (Z_i) ;
- Conditionally on $Z_i = k$, the variable Y_i follows a parametric distribution $F_k = F(\gamma_k)$ with density function $f_k(\cdot) = f(\cdot; \gamma_k)$.

The objective of inferring a mixture model is to estimate the following parameters:

- π_k : proportion of the population belonging to class k ;
- γ_k : parameters of the F_k distribution.

The set of parameters to be estimated is denoted by:

$$\theta := (\pi, \gamma) = ((\pi_k), (\gamma_k)).$$

In practice, mixture models are most often used for classification purposes, so the main objective is to infer the latent status Z_i of each individual.

Definition An **independent mixture model** is defined as follows:

$$\begin{aligned} (Z_i)_i &\text{ i.i.d.,} & Z_i &\sim \mathcal{M}(1; \pi), \\ (Y_i)_i &\text{ indep. } | (Z_i), & Y_i | (Z_i = k) &\sim F_k = F(\gamma_k), \end{aligned}$$

2.2 Latent Block Model (LBM)

Latent block models (Govaert and Nadif (2003)) are useful for identify group structure in a bipartite graph, for example. Latent blocks are separated into two disjoint sets $Q_r = \{1, \dots, Q_r\}$ and $Q_c = \{1, \dots, Q_c\}$. They each have a respective latent variable, Z and W with $(Z_i)_{i \in Q_r}$ and $(W_j)_{j \in Q_c}$ i.i.d such that $Z_i = q$ if and only if node i belongs to block q_r . Similarly for $W_j = l$. Then we have:

$$\begin{aligned} \mathbb{P}(Z_i = q) &= \pi_q, \quad \sum_{q \in Q_r} \pi_q = 1 \\ \mathbb{P}(W_j = l) &= \rho_l, \quad \sum_{l \in Q_c} \rho_l = 1 \end{aligned}$$

where $\pi = (\pi_1, \dots, \pi_{Q_r})$, $\rho = (\rho_1, \dots, \rho_{Q_c})$ are the mixture parameters, i.e. the proportions of individuals in each group.

The probability that two nodes i and j are connected by an edge is:

$$\mathbb{P}(X_{ij} = 1 \mid Z_i = q, W_j = \ell) = \lambda_{q\ell}$$

with $\lambda = (\lambda_{q\ell})_{q \in Q_r \times \ell \in Q_c}$ the inter-block connection matrix.

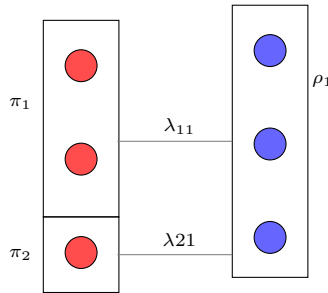


Figure 1: LBM for a bipartite graph

3 Algorithms Theory

In this part, we will show the theoretical implementation, assumptions and convergence of algorithms to infer Latent Block Model. All the proposition are demonstrated in Annexes.

Firstly, we define the parameters of LBM:

$$Y \sim LBM(K, L, \pi_K, \rho_L, \lambda_{KL})$$

and we define the following LBM :

$$M_{KL} = \begin{cases} Z_i \sim M(1, \pi) \\ W_j \sim M(1, \rho) \\ Y_{ij} \mid Z_i, W_j \sim \text{Bern}(Z_i^T \lambda W_j) \end{cases}$$

with Z_i (resp. W_j) the latent variables associated to the rows (resp. columns). $(Z_i)_{1 \leq i \leq n}$ (resp. $(W_j)_{1 \leq j \leq m}$) are independent vectors with K (resp. L) coordinates where all are zero except one. We note $\pi_k (1 \leq k \leq K)$ (resp. $\rho_l (1 \leq l \leq L)$) the probability that the k -th (resp. l -th) coordinate is non zero.

λ is a matrix $K \times L$ with $K, L > 1$ which contains probability of linking between groups k and l

3.1 EM Algorithm

In models like the Latent Block Model, the observed data Y depends on latent variables Z and W , which represent unobserved row and column cluster memberships. Since these latent variables are not observed, we cannot directly compute or maximize the complete-data likelihood.

The Expectation-Maximization (EM) algorithm, introduced by Dempster et al. (1977), provides a general framework to estimate parameters in such latent variable models by iteratively maximizing a lower bound of the marginal log-likelihood.

The goal is to maximize the marginal log-likelihood:

$$\log p(Y|\theta) = \log \sum_Z \sum_W p(Y, Z, W|\theta)$$

However, this expression is intractable due to the summation over all possible configurations of (Z, W) .

To make the problem tractable, we introduce an auxiliary distribution $q(Z, W)$ over the latent variables, and apply Jensen's inequality :

$$\log p(Y|\theta) = \log \sum_{Z, W} q(Z, W) \frac{p(Y, Z, W|\theta)}{q(Z, W)} \geq \sum_{Z, W} q(Z, W) \log \frac{p(Y, Z, W|\theta)}{q(Z, W)} = \mathcal{L}(q, \theta)$$

We define the Evidence Lower Bound (ELBO) as:

$$\mathcal{L}(q, \theta) = \sum_{Z, W} q(Z, W) \log \frac{p(Y, Z, W|\theta)}{q(Z, W)} = \mathbb{E}_q[\log p(Y, Z, W|\theta)] - \mathbb{E}_q[\log q(Z, W)]$$

The EM algorithm can be interpreted as a coordinate ascent on this lower bound $\mathcal{L}(q, \theta)$:

- **E-step:** For fixed $\theta^{(t)}$, choose $q^{(t+1)}(Z, W) = p(Z, W|Y, \theta^{(t)})$. This step tightens the bound and makes it equal to the true log-likelihood at $\theta^{(t)}$.
- **M-step:** Maximize the ELBO with respect to θ :

$$\theta^{(t+1)} = \arg \max_{\theta} \mathbb{E}_{q^{(t+1)}}[\log p(Y, Z, W|\theta)]$$

This procedure guarantees that the likelihood increases at each step:

$$\log p(Y|\theta^{(t+1)}) \geq \log p(Y|\theta^{(t)})$$

until convergence to a (local) maximum.

Proposition 3.1 *The E-step consists of computing the responsibilities (posterior probabilities) defined as:*

$$\eta_{ij,kl}^{(t)} = \mathbb{P}(Z_i = k, W_j = l \mid Y, \theta^{(t)}) = \frac{\sum_{Z: Z_i=k} \sum_{W: W_j=l} p(Y, Z, W \mid \theta^{(t)})}{\sum_Z \sum_W p(Y, Z, W \mid \theta^{(t)})}$$

This quantity represents the posterior probability that row i belongs to cluster k and column j belongs to cluster l , given the observed data and current parameter estimates.

However, this exact computation is intractable, as it involves summing over all possible configurations of the latent variables Z and W . The total number of such configurations is exponential:

$$K^n \cdot L^m$$

Thus, computing each responsibility $\eta_{ij,kl}^{(t)}$ would require evaluating a sum with $K^n L^m$ terms, which is computationally infeasible for large n and m .

To address this, we can do a variational approximation (Variational EM) or a stochastic approximation (Markov Chain Monte Carlo).

3.2 VEM Algorithm

For the VEM algorithm, we introduce a tractable approximation of the posterior using a variational distribution :

$$q(Z, W) = \prod_{i=1}^n q(Z_i) \prod_{j=1}^m q(W_j)$$

where $q(Z_i = k) = \tau_{ik}$ and $q(W_j = l) = \xi_{jl}$. These quantities, known as the variational responsibilities, are updated iteratively to approximate the true posterior in a computationally efficient way.

Proposition 3.2 *The E-step for each distribution $q(Z_i)$ and $q(W_j)$ is given by :*

$$\begin{aligned} q(Z_i) &= M(Z_i; 1, \tau_i) \\ q(W_j) &= M(W_j; 1, \xi_j) \end{aligned}$$

$$\text{where } \sum_{k=1}^K \tau_{ik} = 1, \sum_{l=1}^L \xi_{jl} = 1$$

$$\begin{aligned} \tau_{ik} &\propto \exp \left(\log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})) \right) \\ \xi_{jl} &\propto \exp \left(\log \rho_l + \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})) \right) \end{aligned}$$

Proposition 3.3 *The M-step to update the parameter is given by :*

$$\begin{aligned} \pi_k &= \frac{1}{n} \sum_{i=1}^n \tau_{ik} \\ \rho_l &= \frac{1}{m} \sum_{j=1}^m \xi_{jl} \\ \lambda_{kl} &= \frac{\sum_{i=1}^n \sum_{j=1}^m \tau_{ik} \xi_{jl} Y_{ij}}{\sum_{i=1}^n \sum_{j=1}^m \tau_{ik} \xi_{jl}} \end{aligned}$$

Proposition 3.4 *After the M-step, the ELBO is given by :*

$$\begin{aligned} \mathcal{L}(q, \theta) &= \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \eta_{jl} \log \rho_l + \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L \tau_{ik} \eta_{jl} \log p(Y_{ij} | \lambda_{kl}) \\ &\quad - \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \tau_{ik} - \sum_{j=1}^m \sum_{l=1}^L \eta_{jl} \log \eta_{jl} \end{aligned}$$

With the variational approximation, the complexity becomes $O(nmKL)$, we sum on all couple (i, j) and accumulate on $K \times L$ combinations.

In practice, the VEM algorithm is initialized using k-means clustering applied to both the rows and the columns of the adjacency matrix. Specifically, the rows (respectively columns) of the network are clustered into K (respectively L) groups based on their connection profiles. The resulting hard cluster assignments are then converted into soft probabilistic memberships to initialize the variational parameters τ and ξ . This approach provides a more stable and informed initialization than random assignment, reducing the risk of convergence toward poor local optima.

3.3 Markov Chain Monte Carlo (MCMC) and Gibbs Sampling

In the E-step of EM algorithm, we would compute the posterior distribution $p(Z, W|Y, \theta^{(t)})$. However, this approach faces a fundamental computational barrier: the exact calculation of $p(Z, W|Y, \theta^{(t)})$ is intractable.

Gibbs sampling, introduced by Geman and Geman (1984), which is a particular Markov Chain Monte Carlo (MCMC) methods offer an elegant solution to this intractability by reframing the problem in a Bayesian framework. The algorithm proceeds by iteratively sampling each variable from its full conditional distribution, holding all other variables fixed.

- Sample each row cluster assignment Z_i from $p(Z_i|Z_{-i}, W, \theta, Y)$
- Sample each column cluster assignment W_j from $p(W_j|Z, W_{-j}, \theta, Y)$
- Sample the parameters θ from $p(\theta|Z, W, Y)$

This creates a Markov chain whose stationary distribution is exactly the joint posterior $p(Z, W, \theta|Y)$. After a burn-in period, the samples can be used to approximate posterior expectations, compute parameter estimates and quantify uncertainty.

Each conditional sampling step is computationally tractable, requiring $O(K)$ or $O(L)$ operations per update. In the following section, we derive the specific full conditional distributions for the LBM and describe the implementation of the Gibbs sampler.

We start by defining prior distribution on the parameters. As π and ρ are involved in Multinomial distributions to generate Z_i, W_j , we choose Dirichlet distribution :

$$\begin{aligned}\pi &\sim \text{Dir}(e) \text{ with } e = (e_1, \dots, e_K) \\ \rho &\sim \text{Dir}(f) \text{ with } f = (f_1, \dots, f_L)\end{aligned}$$

Morover, $Y_{ij}|Z_i, W_j, \lambda_{kl}$ follows a Bernoulli distribution, we will use prior distribution Beta for each term of the matrix λ :

$$\lambda_{kl} \sim \text{Beta}(a_{kl}, b_{kl})$$

with $a_{kl} = b_{kl} = 1$ which correspond to non-informative prior.

We can write the joint posterior as :

$$p(Z, W, \pi, \rho, \lambda|Y) = \frac{p(Z, W, \pi, \rho, \lambda, Y)}{p(Y)} \propto p(Y | Z, W, \lambda) p(Z|\pi) p(W|\rho) p(\pi) p(\rho) p(\lambda)$$

We define the Gibbs scheme for a LBM like this :

$$\begin{cases} Z_i^{(t+1)} \sim p(Z_i | Y, W^{(t)}, \lambda^{(t)}, \pi^{(t)}) \\ W_j^{(t+1)} \sim p(W_j | Y, Z^{(t+1)}, \lambda^{(t)}, \rho^{(t)}) \\ \lambda_{k\ell}^{(t+1)} \sim p(\lambda_{k\ell} | Y, Z^{(t+1)}, W^{(t+1)}) \\ \pi^{(t+1)} \sim p(\pi | Z^{(t+1)}) \\ \rho^{(t+1)} \sim p(\rho | W^{(t+1)}) \end{cases}$$

Proposition 3.5 *The update of π and ρ is given by :*

$$\begin{aligned}\pi|Z &\sim \text{Dir}(n_1 + e_1, \dots, n_K + e_K) \text{ with } n_k = \sum_{i=1}^n \mathbb{1}_{\{Z_i=k\}} \\ \rho|W &\sim \text{Dir}(m_1 + f_1, \dots, m_L + f_L) \text{ with } m_l = \sum_{j=1}^m \mathbb{1}_{\{W_j=l\}}\end{aligned}$$

Proposition 3.6 *The update of λ is given by :*

$$\lambda_{kl}|Y, Z, W \sim \text{Beta}(a_{kl} + S_{kl}, b_{kl} + N_{kl} - S_{kl}),$$

this is a Beta-Bernoulli distribution and

$$S_{kl} = \sum_{i:Z_i=k} \sum_{j:W_j=l} Y_{ij} \text{ (number of 1 in block } (k, l))$$

$$N_{kl} = (\#\{i : Z_i = k\})(\#\{j : W_j = l\}) \text{ (size of block)}$$

Proposition 3.7 *The update of each Z_i and W_j is given by :*

$$p(Z_i = k|-) \propto \pi_k \prod_{j=1}^m \prod_{l=1}^L \left[(\lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}})^{\mathbb{1}_{\{W_j=l\}}}\right]$$

$$p(W_j = l|-) \propto \rho_l \prod_{i=1}^n \prod_{k=1}^K \left[(\lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}})^{\mathbb{1}_{\{Z_i=k\}}}\right]$$

Proposition 3.8 *The Markov Chain generated by a Gibbs Sampler is irreducible, aperiodic and Harris recurrence. Then, following the distribution convergence theorem for a markov chain, the Gibbs Sampler for a LBM converge :*

$$\lim_{t \rightarrow \infty} \mathcal{L}(Z^{(t)}, W^{(t)}, \theta^{(t)}) = p(Z, W, \theta|Y)$$

In practice, the initial iterations of the Gibbs sampler, called the burn-in period, are discarded because the Markov chain has not yet converged to its stationary distribution and still depends on the starting values. After the burn-in, samples are collected from the chain. To reduce autocorrelation between successive samples, a technique called thinning is often applied, where only every k -th sample is retained. Finally, model parameters such as π , ρ , λ are estimated by averaging the retained samples, providing stable posterior estimates.

4 Results

In this section, we present our implementation of the Gibbs Sampler and the Variational EM (VEM) algorithm (on this git), and we compare their empirical behaviour with the theoretical results discussed earlier. Both algorithms have been implemented in R. As previously mentioned, the standard EM algorithm is expected to fail to converge or to require prohibitive computation time in the case of the Latent Block Model (LBM).

We first describe how we generate synthetic data from a random Latent Block Model, as well as the simulation protocol used for testing. We then introduce the evaluation metric of interest, namely the Adjusted Rand Index (ARI). The objective is to assess the convergence behaviour and estimation accuracy of both inference methods, and to identify in which regimes (network size, number of blocks) each algorithm performs best. Finally, we apply both methods to empirical prey-predator networks from the Balearic Islands, in order to assess their performance on real ecological data.

4.1 Simulation of LBM

To generate synthetic datasets, we simulate a bipartite network according to a Latent Block Model (LBM). The simulation follows the steps below:

1. Choose the model parameters:

- Fix the number of row clusters $K > 1$ and column clusters $L > 1$
- **Cluster membership probabilities :** The proportions of nodes in each latent group are drawn from symmetric Dirichlet distributions: $\pi \sim \text{Dirichlet}(\mathbf{1}_K)$ and $\rho \sim \text{Dirichlet}(\mathbf{1}_L)$, where $\mathbf{1}_K$ and $\mathbf{1}_L$ denote K - and L -dimensional vectors of ones. This choice corresponds to a non-informative prior, meaning that each cluster has initially the same expected weight, $\mathbb{E}[\pi_k] = 1/K$ and $\mathbb{E}[\rho_l] = 1/L$.

- **Connectivity matrix** : The connectivity between row and column clusters is encoded in a $K \times L$ matrix $\lambda = (\lambda_{kl})$, where each entry represents the probability of an edge between a node from cluster k (rows) and a node from cluster l (columns). To simulate heterogeneous interaction patterns, each λ_{kl} is sampled independently from a uniform distribution:

$$\lambda_{kl} \sim \text{Uniform}(\lambda_{\min}, \lambda_{\max}), \text{ with } \lambda_{\min} = 0.01, \lambda_{\max} = 0.99$$

This ensures variability in connection probabilities across block pairs while keeping all values within $[\lambda_{\min}, \lambda_{\max}]$.

2. Generate cluster memberships:

- For each row node $i = 1, \dots, n$, draw its cluster assignment $Z_i \sim \mathcal{M}(1, \pi)$.
- For each column node $j = 1, \dots, m$, draw its cluster assignment $W_j \sim \mathcal{M}(1, \rho)$.

3. Generate the bipartite adjacency matrix:

- For each pair (i, j) , sample the interaction according to a Bernoulli distribution:

$$Y_{ij} \sim \text{Bernoulli}(Z_i^\top \lambda W_j).$$

In practice, this procedure is implemented in R using nested loops: the row and column cluster labels Z and W are first drawn from their respective multinomial distributions, and then each entry Y_{ij} of the adjacency matrix is generated from a Bernoulli distribution with success probability $\lambda_{Z_i W_j}$.

4.2 Evaluation Metric

Adjusted Rand Index (ARI) To evaluate the accuracy of the estimated blocks assignments, we use the Adjusted Rand Index (ARI), a classical measure of agreement between two partitions developed by (Hubert and Arabie, 1985).

Let n be the number of observations and let $U = \{U_1, \dots, U_K\}$ and $V = \{V_1, \dots, V_L\}$ denote two partitions (the true and the estimated clusterings in our case). Let n_{kl} be the number of elements jointly assigned to cluster U_k and V_l , $a_k = \sum_l n_{kl}$ the size of cluster U_k , and $b_l = \sum_k n_{kl}$ the size of cluster V_l . The Adjusted Rand Index is defined as:

$$\text{ARI} = \frac{\sum_{k,l} \binom{n_{kl}}{2} - \frac{\sum_k \binom{a_k}{2} \sum_l \binom{b_l}{2}}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_k \binom{a_k}{2} + \sum_l \binom{b_l}{2} \right] - \frac{\sum_k \binom{a_k}{2} \sum_l \binom{b_l}{2}}{\binom{n}{2}}}$$

The first term $\sum_{k,l} \binom{n_{kl}}{2}$ counts the number of pairs of elements that are consistently assigned to the same cluster in both partitions, while the denominator adjusts for agreements expected by chance.

The ARI takes values in the interval $[-1, 1]$: it equals 1 when the two partitions are identical, 0 when the similarity corresponds to random labeling, and may become negative when the agreement is worse than random.

In the context of the Latent Block Model (LBM), we compute the ARI separately for the row and column clusterings, which allows for a clear comparison of the convergence and accuracy of inference algorithms such as VEM and MCMC.

4.3 Test on Synthetic Dataset

We use our algorithms on LBMs generated with different combinations on $n * m$ and $K * L$, respectively $n, m \in \{10, 50, 100, 200\}$ and $K, L \in \{2, 4, 8\}$. The idea is to identify how the

variation of these parameters influences the convergence of both algorithms but also see in which combinations of parameters each algorithms performs best.

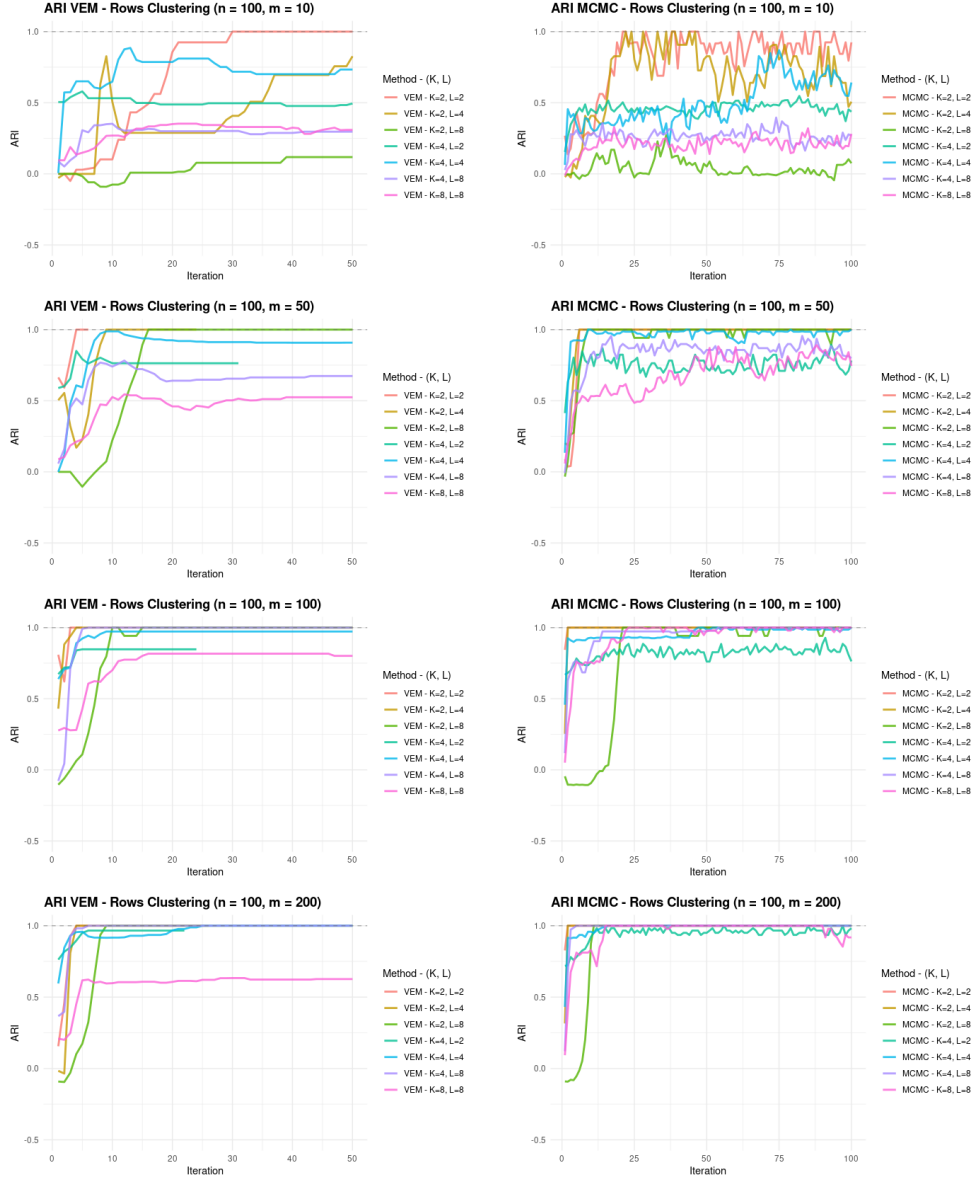


Figure 2: ARI rows comparison for a network of size $100 \times m$

The figure 2 shows the evolution of the Adjusted Rand Index (ARI) for both the MCMC (Gibbs sampler) and the Variational EM (VEM) algorithms, for different network sizes ($n = 100, m \in \{10, 50, 100, 200\}$).

We observe that the larger the network, the better the convergence for both methods, as the increased amount of data provides more information to correctly infer the latent structure. The convergence of the MCMC appears generally smoother and more stable, whereas the VEM can sometimes get trapped in local maxima and therefore exhibits poorer convergence. This behavior can be explained by the deterministic nature of VEM, which tends to stay in a local optimum once reached, while MCMC, through its stochastic sampling, can explore the parameter space more effectively and occasionally escape suboptimal regions. Additionally, when the number of latent groups increases, VEM tends to

struggle more with convergence, whereas MCMC mainly faces difficulties when there is an imbalance between the number of row and column clusters.

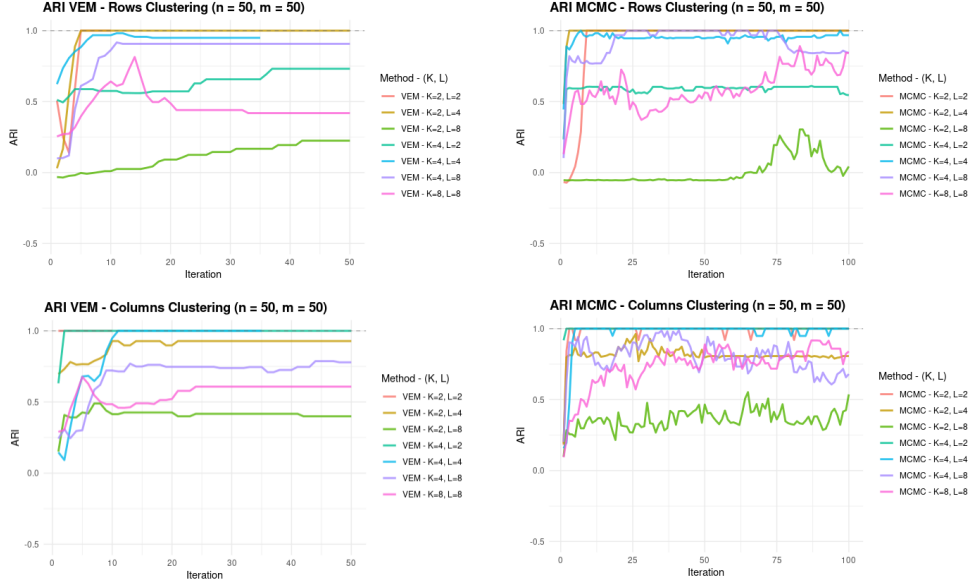


Figure 3: ARI rows and columns for a network of size 50×50

For networks of size 50×50 on 3, we display the evolution of the Adjusted Rand Index (ARI) for both row and column clusterings under the MCMC and VEM algorithms. The convergence patterns of the row and column ARIs are generally similar across methods. However, both algorithms tend to struggle when the numbers of row and column clusters are unbalanced for example, when $K \neq L$. In such cases, convergence is poorer and the algorithms may become trapped in suboptimal configurations, leading to less stable ARI values over iterations.

4.4 Test on Real Dataset

We inferred four LBMs from ecological networks from the Balearic Islands, including two predator–prey networks and two plant–herbivore networks. For each type of network, one was collected from a disturbed island while the other was obtained from a pristine island. In these networks, the rows represent plants or prey species, and the columns represent herbivores or predators. To obtain LBM on bipartite network, we have used the package *sbm* in R.



Figure 4: ARI for ecological network

The VEM algorithm performs slightly better than the MCMC in terms of ARI convergence. However, neither method reaches perfect agreement ($ARI = 1$) for the plant clustering in the pristine island network. This lack of convergence is mainly due to the unbalanced block structure of the network. In particular, for herbivores, one of the groups contains only a single individual while the other includes around ten, making the estimation of group assignments more unstable.

5 Conclusion

Our comparative study of VEM and MCMC (Gibbs sampling) reveals distinct trade-offs between these approaches. Theoretically, VEM uses a deterministic variational approximation with complexity $O(nmKL)$, making it computationally efficient but potentially trapped in local optima. MCMC provides asymptotic convergence guarantees to the true posterior through stochastic sampling, but requires careful tuning of burn-in and thinning parameters.

Empirically, our simulations on synthetic data show that both algorithms improve with larger network sizes. However, VEM converges faster but struggles when the number of clusters increases or when $K \neq L$. MCMC exhibits smoother, more stable convergence and better explores the parameter space, though it is slower per iteration. On real ecological networks from the Balearic Islands, VEM achieved slightly faster ARI convergence to higher ARI scores, but both methods faced difficulties with highly unbalanced block structures.

In conclusion, the choice between VEM and MCMC depends on the application context. VEM is preferable for computational efficiency and balanced structures, while MCMC is superior for uncertainty quantification and robust exploration when multiple local optima exist. Neither algorithm dominates universally, and the optimal choice requires considering the network’s structural properties and inferential objectives.

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Appendix

A Proof of Expectation Maximization

A.1 Computation of E-step

We want to compute $p(Z, W|Y, \theta^{(t)})$. We introduce the responsibilities:

$$\begin{aligned}\gamma_{ik}^{(t)} &= p(Z_i = k|Y, \theta^{(t)}) \\ \delta_{jl}^{(t)} &= p(W_j = l|Y, \theta^{(t)}) \\ \eta_{ij,kl}^{(t)} &= p(Z_i = k, W_j = l|Y, \theta^{(t)})\end{aligned}$$

The E-step of the EM algorithm requires computing the expected complete-data log-likelihood:

$$\begin{aligned}Q(\theta|\theta^{(t)}) &= \mathbb{E}_{Z, W|Y, \theta^{(t)}}[\log p(Y, Z, W|\theta)] \\ &= \sum_Z \sum_W p(Z, W|Y, \theta^{(t)}) \log p(Y, Z, W|\theta)\end{aligned}$$

Using the LBM structure, the complete-data log-likelihood is:

$$\begin{aligned}\log p(Y, Z, W|\theta) &= \sum_{i=1}^n \sum_{k=1}^K \mathbb{1}_{\{Z_i=k\}} \log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \mathbb{1}_{\{W_j=l\}} \log \rho_l \\ &\quad + \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L \mathbb{1}_{\{Z_i=k, W_j=l\}} [Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})]\end{aligned}$$

Taking the expectation with respect to $p(Z, W|Y, \theta^{(t)})$ and replacing the indicators by their expectations (the responsibilities):

$$\begin{aligned} Q(\theta|\theta^{(t)}) &= \sum_{i=1}^n \sum_{k=1}^K \gamma_{ik}^{(t)} \log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \delta_{jl}^{(t)} \log \rho_l \\ &\quad + \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L \eta_{ij,kl}^{(t)} [Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})] \end{aligned}$$

To compute these responsibilities, we need to evaluate the joint posterior:

$$\eta_{ij,kl}^{(t)} = p(Z_i = k, W_j = l|Y, \theta^{(t)}) = \sum_{Z: Z_i=k} \sum_{W: W_j=l} p(Z, W|Y, \theta^{(t)})$$

By Bayes theorem:

$$p(Z, W|Y, \theta^{(t)}) = \frac{p(Y, Z, W|\theta^{(t)})}{p(Y|\theta^{(t)})}$$

where the marginal likelihood is:

$$p(Y|\theta^{(t)}) = \sum_Z \sum_W p(Y, Z, W|\theta^{(t)})$$

Therefore:

$$\eta_{ij,kl}^{(t)} = \frac{\sum_{Z: Z_i=k} \sum_{W: W_j=l} p(Y, Z, W|\theta^{(t)})}{\sum_Z \sum_W p(Y, Z, W|\theta^{(t)})}$$

B Proof of Variational EM

B.1 Computation of E-step

In the variational EM framework, we approximate the intractable posterior $p(Z, W|Y, \theta)$ with a factorized distribution:

$$q(Z, W) = \prod_{i=1}^n q(Z_i) \prod_{j=1}^m q(W_j)$$

The optimal variational distribution satisfies:

$$\log q(Z_i) = \mathbb{E}_{q(W)}[\log p(Y|Z_i, W, \lambda)] + \mathbb{E}_{q(\pi)}[\log p(Z_i|\pi)] + \text{const}$$

Since $Y_{ij}|Z_i, W_j \sim \text{Bern}(Z_i^T \lambda W_j)$, we have:

$$\begin{aligned} p(Y_{ij}|Z_i, W_j, \lambda) &= \lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}} \text{ when } Z_{ik} = 1, W_{jl} = 1 \\ \log p(Y_{ij}|Z_i, W_j, \lambda) &= Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl}) \end{aligned}$$

Fixing i and summing over all possible combinations of k, l, j :

$$\log p(Y_i|Z_i, W, \lambda) = \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L Z_{ik} W_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl}))$$

Taking the expectation with respect to $q(W)$, where $q(W_j = l) = \xi_{jl}$:

$$\mathbb{E}_{q(W)}[\log p(Y_i|Z_i, W, \lambda)] = \sum_{k=1}^K Z_{ik} \sum_{j=1}^m \sum_{l=1}^L \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl}))$$

For the prior term:

$$p(Z_i|\pi) = \prod_{k=1}^K \pi_k^{Z_{ik}}$$

$$\log p(Z_i|\pi) = \sum_{k=1}^K Z_{ik} \log \pi_k$$

Combining both terms:

$$\log q(Z_i) = \sum_{k=1}^K Z_{ik} \left(\sum_{j=1}^m \sum_{l=1}^L \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})) + \log \pi_k \right) + \text{const}$$

Therefore, $q(Z_i) = \mathcal{M}(Z_i; 1, \tau_i)$ where $\sum_{k=1}^K \tau_{ik} = 1$ and:

$$\tau_{ik} \propto \exp \left(\sum_{j=1}^m \sum_{l=1}^L \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})) + \log \pi_k \right)$$

By symmetry, proceeding with the same computation for $q(W_j)$:

B.2 Update for π and ρ

Maximizing $\sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \pi_k$ subject to $\sum_k \pi_k = 1$:

$$\pi_k = \frac{1}{n} \sum_{i=1}^n \tau_{ik}$$

Similarly:

$$\rho_l = \frac{1}{m} \sum_{j=1}^m \xi_{jl}$$

B.3 Update for λ

Maximizing:

$$\sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L \tau_{ik} \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl}))$$

Taking the derivative with respect to λ_{kl} and setting it to zero:

$$\frac{\partial}{\partial \lambda_{kl}} = \sum_{i=1}^n \sum_{j=1}^m \tau_{ik} \xi_{jl} \left(\frac{Y_{ij}}{\lambda_{kl}} - \frac{1 - Y_{ij}}{1 - \lambda_{kl}} \right) = 0$$

Solving for λ_{kl} :

$$\lambda_{kl} = \frac{\sum_{i=1}^n \sum_{j=1}^m \tau_{ik} \xi_{jl} Y_{ij}}{\sum_{i=1}^n \sum_{j=1}^m \tau_{ik} \xi_{jl}}$$

B.4 Computation of ELBO

The Evidence Lower BOund (ELBO) is defined as:

$$\mathcal{L}(q, \theta) = \mathbb{E}_q[\log p(Y, Z, W|\theta)] - \mathbb{E}_q[\log q(Z, W)]$$

$$\begin{aligned} \mathbb{E}_q[\log p(Y, Z, W|\theta)] &= \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \xi_{jl} \log \rho_l \\ &\quad + \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L \tau_{ik} \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})) \end{aligned}$$

Since $q(Z, W) = \prod_{i=1}^n q(Z_i) \prod_{j=1}^m q(W_j)$:

$$\mathbb{E}_q[\log q(Z, W)] = \sum_{i=1}^n \mathbb{E}_q[\log q(Z_i)] + \sum_{j=1}^m \mathbb{E}_q[\log q(W_j)]$$

For $q(Z_i) = \mathcal{M}(Z_i; 1, \tau_i)$ and $q(W_j) = \mathcal{M}(W_j; 1, \xi_j)$:

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}_q[\log q(Z_i)] &= \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \tau_{ik} \\ \sum_{j=1}^m \mathbb{E}_q[\log q(W_j)] &= \sum_{j=1}^m \sum_{l=1}^L \xi_{jl} \log \xi_{jl} \end{aligned}$$

Combining all terms:

$$\begin{aligned} \mathcal{L}(q, \theta) &= \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \xi_{jl} \log \rho_l \\ &\quad + \sum_{i=1}^n \sum_{j=1}^m \sum_{k=1}^K \sum_{l=1}^L \tau_{ik} \xi_{jl} (Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})) \\ &\quad - \sum_{i=1}^n \sum_{k=1}^K \tau_{ik} \log \tau_{ik} - \sum_{j=1}^m \sum_{l=1}^L \xi_{jl} \log \xi_{jl} \end{aligned}$$

C Proof of Gibbs Sampler

C.1 Update of π, ρ

By Bayes theorem:

$$p(\pi|Z) \propto p(Z|\pi)p(\pi)$$

The likelihood term is:

$$p(Z|\pi) = \prod_{i=1}^n p(Z_i|\pi) = \prod_{i=1}^n \pi_{Z_i} = \prod_{k=1}^K \pi_k^{n_k}$$

where $n_k = \sum_{i=1}^n \mathbb{1}_{\{Z_i=k\}}$ is the number of rows assigned to cluster k .

Since we chose the prior $\pi \sim \text{Dir}(e_1, \dots, e_K)$, we have:

$$p(\pi) \propto \prod_{k=1}^K \pi_k^{e_k-1}$$

Combining the likelihood and prior:

$$\begin{aligned}
p(\pi|Z) &\propto p(Z|\pi) \cdot p(\pi) \\
&= \prod_{k=1}^K \pi_k^{n_k} \cdot \prod_{k=1}^K \pi_k^{e_k-1} \\
&= \prod_{k=1}^K \pi_k^{n_k+e_k-1}
\end{aligned}$$

This is the kernel of a Dirichlet distribution, therefore:

$$\pi|Z \sim \text{Dir}(n_1 + e_1, \dots, n_K + e_K)$$

By symmetry, proceeding in the same way for ρ with prior $\rho \sim \text{Dir}(f_1, \dots, f_L)$:

$$\rho|W \sim \text{Dir}(m_1 + f_1, \dots, m_L + f_L)$$

where $m_l = \sum_{j=1}^m \mathbb{1}_{\{W_j=l\}}$ is the number of columns assigned to cluster l .

C.2 Update of λ

By Bayes theorem:

$$p(\lambda_{kl}|Y, Z, W) \propto p(Y|Z, W, \lambda_{kl}) \cdot p(\lambda_{kl})$$

For a block pair (k, l) , we define:

$$\begin{aligned}
S_{kl} &= \sum_{i:Z_i=k} \sum_{j:W_j=l} Y_{ij} \quad (\text{number of edges in block } (k, l)) \\
N_{kl} &= (\#\{i : Z_i = k\}) \cdot (\#\{j : W_j = l\}) \quad (\text{size of block } (k, l))
\end{aligned}$$

The contribution of block (k, l) to the likelihood is:

$$\begin{aligned}
p(Y_{(k,l)}|Z, W, \lambda_{kl}) &= \prod_{i:Z_i=k} \prod_{j:W_j=l} p(Y_{ij}|\lambda_{kl}) \\
&= \prod_{i:Z_i=k} \prod_{j:W_j=l} \lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}} \\
&= \lambda_{kl}^{S_{kl}} (1 - \lambda_{kl})^{N_{kl}-S_{kl}}
\end{aligned}$$

This follows because:

- λ_{kl} appears S_{kl} times (once for each $Y_{ij} = 1$ in block (k, l))
- $(1 - \lambda_{kl})$ appears $N_{kl} - S_{kl}$ times (once for each $Y_{ij} = 0$ in block (k, l))

Since we chose the prior $\lambda_{kl} \sim \text{Beta}(a_{kl}, b_{kl})$:

$$p(\lambda_{kl}) \propto \lambda_{kl}^{a_{kl}-1} (1 - \lambda_{kl})^{b_{kl}-1}$$

Combining the likelihood and prior:

$$\begin{aligned}
p(\lambda_{kl}|Y, Z, W) &\propto p(Y|Z, W, \lambda_{kl}) \cdot p(\lambda_{kl}) \\
&= \lambda_{kl}^{S_{kl}} (1 - \lambda_{kl})^{N_{kl}-S_{kl}} \cdot \lambda_{kl}^{a_{kl}-1} (1 - \lambda_{kl})^{b_{kl}-1} \\
&= \lambda_{kl}^{S_{kl}+a_{kl}-1} (1 - \lambda_{kl})^{N_{kl}-S_{kl}+b_{kl}-1}
\end{aligned}$$

This is the kernel of a Beta distribution, therefore:

$$\lambda_{kl}|Y, Z, W \sim \text{Beta}(a_{kl} + S_{kl}, b_{kl} + N_{kl} - S_{kl})$$

C.3 Update of Z_i, W_j

By Bayes theorem:

$$p(Z_i = k|Y, Z_{-i}, W, \pi, \rho, \lambda) \propto p(Z_i = k, Y|Z_{-i}, W, \pi, \rho, \lambda) \times p(Z_i = k|\pi)$$

We can factorize the joint distribution using the conditional independence structure of the LBM:

$$p(Z_i = k, Y|Z_{-i}, W, \pi, \rho, \lambda) = p(Y|Z_i = k, Z_{-i}, W, \lambda) \times p(Z_i = k|\pi)$$

Since the observations Y_{ij} are conditionally independent given the cluster assignments and parameters, and row i only affects observations in row i :

$$\begin{aligned} p(Y|Z_i = k, Z_{-i}, W, \lambda) &= p(Y_{i\cdot}|Z_i = k, W, \lambda) \times p(Y_{-i\cdot}|Z_{-i}, W, \lambda) \\ &\propto p(Y_{i\cdot}|Z_i = k, W, \lambda) \end{aligned}$$

where $Y_{i\cdot} = (Y_{i1}, \dots, Y_{im})$ denotes the i -th row of the observation matrix.

For each observation Y_{ij} in row i :

$$Y_{ij}|Z_i = k, W_j = l, \lambda_{kl} \sim \text{Bernoulli}(\lambda_{kl})$$

Therefore:

$$\begin{aligned} p(Y_{i\cdot}|Z_i = k, W, \lambda) &= \prod_{j=1}^m p(Y_{ij}|Z_i = k, W_j, \lambda) \\ &= \prod_{j=1}^m \sum_{l=1}^L \mathbb{1}_{\{W_j=l\}} p(Y_{ij}|Z_i = k, W_j = l, \lambda_{kl}) \\ &= \prod_{j=1}^m \sum_{l=1}^L \mathbb{1}_{\{W_j=l\}} \lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}} \\ &= \prod_{j=1}^m \prod_{l=1}^L \left[\lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}} \right]^{\mathbb{1}_{\{W_j=l\}}} \end{aligned}$$

The prior distribution for Z_i is:

$$p(Z_i = k|\pi) = \pi_k$$

Combining these terms, we obtain:

$$\begin{aligned} p(Z_i = k|Y, Z_{-i}, W, \pi, \rho, \lambda) &\propto p(Y_{i\cdot}|Z_i = k, W, \lambda) \times p(Z_i = k|\pi) \\ &\propto \pi_k \prod_{j=1}^m \prod_{l=1}^L \left[\lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}} \right]^{\mathbb{1}_{\{W_j=l\}}} \end{aligned}$$

Taking the logarithm for numerical stability:

$$\log p(Z_i = k|Y, Z_{-i}, W, \pi, \rho, \lambda) \propto \log \pi_k + \sum_{j=1}^m \sum_{l=1}^L \mathbb{1}_{\{W_j=l\}} [Y_{ij} \log \lambda_{kl} + (1 - Y_{ij}) \log(1 - \lambda_{kl})]$$

To obtain the full conditional distribution, we normalize over all possible values of k :

$$p(Z_i = k|Y, Z_{-i}, W, \pi, \rho, \lambda) = \frac{\pi_k \prod_{j=1}^m \prod_{l=1}^L \left[\lambda_{kl}^{Y_{ij}} (1 - \lambda_{kl})^{1-Y_{ij}} \right]^{\mathbb{1}_{\{W_j=l\}}}}{\sum_{k'=1}^K \pi_{k'} \prod_{j=1}^m \prod_{l=1}^L \left[\lambda_{k'l}^{Y_{ij}} (1 - \lambda_{k'l})^{1-Y_{ij}} \right]^{\mathbb{1}_{\{W_j=l\}}}}$$

Algorithm: To sample $Z_i^{(t+1)}$ at iteration $t + 1$:

1. For each $k \in \{1, \dots, K\}$, compute the unnormalized probability:

$$u_k = \pi_k^{(t)} \prod_{j=1}^m (\lambda_{k, W_j^{(t)}})^{Y_{ij}} (1 - \lambda_{k, W_j^{(t)}})^{1 - Y_{ij}}$$

2. Normalize: $p_k = \frac{u_k}{\sum_{k'=1}^K u_{k'}}$

3. Sample $Z_i^{(t+1)} \sim M(1, (p_1, \dots, p_K))$

By symmetry, the update for W_j follows the same reasoning

C.4 Convergence of Gibbs Sampler

We will prove that the Gibbs Sampler converges for a Latent Block Model. It creates a Markov chain on $S = \{1, \dots, K\}^n \times \{1, \dots, L\}^m \times \Theta$.

Firstly, we denote $P(s \rightarrow s')$, the probability of transition from the state s to the state s' . A Markov chain converges towards a stationary distribution if it satisfies the following assumptions:

1. Irreducibility ($\forall (s, s'), P(s \rightarrow s') > 0$)
2. Aperiodicity (no regular cycle, $P(s \rightarrow s) > 0$)
3. Harris recurrence (because we work on a mixed discrete-continuous space)

Irreducibility Firstly, we show that each configuration (Z, W, θ) has a non-zero probability. The prior distribution is:

$$p(Z, W, \theta) = p(\pi)p(\rho)p(\lambda)p(Z|\pi)p(W|\rho)$$

with

$$\begin{aligned} \pi &\sim \text{Dir}(e_1, \dots, e_K), e_k > 0 \forall k \\ \rho &\sim \text{Dir}(f_1, \dots, f_L), f_l > 0 \forall l \\ \lambda_{kl} &\sim \text{Beta}(a_{kl}, b_{kl}), a_{kl}, b_{kl} > 0 \forall (k, l) \end{aligned}$$

Since the Beta distribution has an open support ($\{x : p(x) > 0\}$) on $(0, 1)$ and the Dirichlet distribution has an open support on the simplex $\{(\pi_1, \dots, \pi_K) : \pi_k > 0, \sum_k \pi_k = 1\}$, therefore $p(Z, W, \theta) > 0, \forall Z, W, \theta$.

Now we compute the probability of transition:

$$p(Z_i = k | -) \propto \pi_k \prod_{j=1}^m \lambda_{k, W_j}^{Y_{ij}} (1 - \lambda_{k, W_j})^{1 - Y_{ij}} \text{ with } \pi_k > 0, \lambda_{kl} \in (0, 1)$$

Therefore $p(Z_i = k | -) > 0$. The same holds for $p(W_j = l | -) > 0$. Moreover, since the Beta and Dirichlet distributions have strictly positive densities on their supports, we can conclude that the Markov chain is irreducible.

Aperiodicity For Z_i :

$$p(Z_i^{(t+1)} = Z_i^{(t)} | -) = \frac{\pi_{Z_i^{(t)}} \prod_{j=1}^m \lambda_{Z_i^{(t)}, W_j}^{Y_{ij}} (1 - \lambda_{Z_i^{(t)}, W_j})^{1 - Y_{ij}}}{\sum_k \pi_k \prod_{j=1}^m \lambda_{k, W_j}^{Y_{ij}} (1 - \lambda_{k, W_j})^{1 - Y_{ij}}} > 0$$

The same holds for W_j . For the parameters θ , their distributions have continuous densities with non-zero probability to stay near the current values. Therefore, the Markov chain is aperiodic.

Harris recurrence Consider a transition $Z_i : k \rightarrow k'$ while all other variables are fixed at s_{-i} . Let $s = (Z_i = k, s_{-i})$ and $s' = (Z_i = k', s_{-i})$.

We denote $p(Z, W, \theta|Y) = h(s_{-i}) \times g(Z_i, s_{-i})$ where $h(s_{-i})$ does not depend on Z_i while $g(Z_i, s_{-i})$ does. Also, let $\pi(\cdot) = p(\cdot|Y)$ denote the posterior distribution.

$$\begin{aligned} p(Z_i = k|s_{-i}, Y) &= \frac{p(Z_i = k, s_{-i}|Y)}{p(s_{-i}|Y)} \\ &= \frac{g(k, s_{-i})h(s_{-i})}{\sum_{k'} g(k', s_{-i})h(s_{-i})} \\ &= \frac{g(k, s_{-i})}{\sum_{k'} g(k', s_{-i})} \end{aligned}$$

Therefore:

$$\frac{p(Z_i = k|s_{-i}, Y)}{p(Z_i = k'|s_{-i}, Y)} = \frac{g(k, s_{-i})}{g(k', s_{-i})} = \frac{\pi(s)}{\pi(s')}$$

Since the transition probability satisfies:

$$\frac{P(s \rightarrow s')}{P(s' \rightarrow s)} = \frac{p(Z_i = k'|s_{-i}, Y)}{p(Z_i = k|s_{-i}, Y)} = \frac{\pi(s')}{\pi(s)}$$

We obtain the detailed balance equation:

$$\pi(s)P(s \rightarrow s') = \pi(s')P(s' \rightarrow s).$$

The same procedure applies for W_j and the parameter updates. Since the state space is a product of discrete sets (for Z and W) and continuous compact sets (for the parameters θ in the simplex and $[0, 1]^{KL}$), and all transition densities are strictly positive on their supports, the chain is Harris recurrent.

As the Gibbs Sampler satisfies irreducibility, aperiodicity, and Harris recurrence, convergence to the stationary distribution $\pi(\cdot|Y)$ is guaranteed by the convergence theorem for Markov chains:

$$\lim_{t \rightarrow \infty} \mathcal{L}(Z^{(t)}, W^{(t)}, \theta^{(t)}) = p(Z, W, \theta|Y)$$